BareMinimum Healthcare Metrics and Implementation Guide

Metric definitions, evidence, data handling, and implementation steps

Purpose and scope

This guide explains the healthcare wellness and sustainability dashboard: what each number means, whether it is entered or calculated, what evidence supports comparisons, and how the application stores and protects records. The platform covers SDG 3 personal wellness, SDG 8 healthcare worker well-being, and SDG 12 healthcare resource and waste monitoring. Its personal records are dated and self-reported unless a future provider connection is explicitly added.

BareMinimum is a wellness and monitoring tool. It is not a medical device and does not diagnose or treat. Workload, safety, waste, and sustainability metrics are operational monitoring inputs; they do not create medical or regulatory determinations.

How to read the dashboard

The SDG overview reports record coverage: non-empty fields in each dataset's latest entry divided by all configured fields in that SDG group, rounded to a whole percent. It also counts dated entries. This measures how much has been recorded, not health, worker safety, sustainability performance, or SDG achievement.

- Recorded inputs are values typed into dated forms. They are labelled self-reported or manual and retain their entry timestamp.
- Calculated indicators use the recorded values and formulas described below. They are not independent measurements or clinical outcomes.
- Guidance comparisons come only from the named public-health sources. Activity targets are for adults; sleep comparisons are general adult guidance. Age and individual circumstances can change what applies.
- The amber Sample example is a separate, read-only presentation fixture. It is never placed in personal history, local storage, or MongoDB. The legacy dashboard tiles are illustrative sample previews and do not sync to MongoDB.
- When server storage is unavailable or the person is not authenticated, new entries remain local to that browser and are labelled local-only. Private account records are stored by the verified server identity.

Metric catalogue

The ranges below are input validation limits for this application, not recommended health or workplace limits. Empty fields are omitted from a saved record; a person must enter at least one value. Numeric values are validated in the browser and again on the server. Text fields are length-limited on the server.

SDG 3 Body measurements

Recorded metric	Meaning and source	Validation
Height	Measured value, entered by the user in centimetres.	0-250 cm
Weight	Measured value, entered by the user in kilograms.	0-500 kg
Waist circumference	Measured value, entered by the user in centimetres. No threshold is assigned by this app.	0-300 cm

SDG 3 Physical activity

Recorded metric	Meaning and source	Validation
Moderate activity	User-reported moderate-intensity minutes for the week.	0-1,000 min/week
Vigorous activity	User-reported vigorous-intensity minutes for the week.	0-1,000 min/week
Strength training	User-reported days with muscle-strengthening activity in the week.	0-7 days/week
Steps	Manual step count; no step recommendation is applied.	0-100,000 steps/day
Active time	Manual daily active minutes; the app does not infer intensity.	0-1,440 min/day
Sedentary time	Manual daily sedentary minutes.	0-1,440 min/day

SDG 3 Hydration

Recorded metric	Meaning and source	Validation
Water consumed	User-entered daily intake in millilitres.	0-20,000 ml/day
Estimated daily target	Optional user-entered estimate. The app supplies no universal fluid requirement.	0-20,000 ml/day

SDG 3 Sleep

Recorded metric	Meaning and source	Validation
Sleep duration	User-reported sleep duration. The current form compares it with general adult guidance but does not infer sleep quality or a disorder.	0-24 hours

SDG 3 Wearable-ready manual fields

Recorded metric	Meaning and source	Validation
Steps	Manually entered value; no wearable provider is connected.	0-100,000 steps
Heart rate	Manually entered beats per minute; not device-verified.	0-250 bpm
Active calories	Manually entered estimate; source and method are not verified.	0-10,000 kcal
Exercise duration	Manually entered duration.	0-1,440 minutes
Sleep duration	Manually entered duration.	0-24 hours

SDG 8 Healthcare worker well-being

Recorded metric	Meaning and source	Validation
Working hours	User-entered hours for the reporting period; period should be understood by the reporting team.	0-168 hours
Shift duration	Reported length of a shift.	0-24 hours
Consecutive shifts	Reported number of consecutive shifts.	0-31 shifts
Breaks taken	Reported break minutes.	0-1,440 minutes
Overtime	Reported overtime hours.	0-168 hours
Patients per shift	Reported workload count.	0-1,000 patients
Safety incidents	Reported safety incident count.	0-10,000 incidents
Needle-stick incidents	Reported needle-stick incident count.	0-10,000 incidents
Workplace injuries	Reported workplace injury count.	0-10,000 incidents
Safety training completion	Reported completion percentage, entered by the user.	0-100%
Wellness/fatigue assessment	Self-rating from 0 to 10. No diagnostic interpretation or risk category is assigned.	0-10

SDG 12 Biomedical waste

Recorded metric	Meaning and source	Validation
Total waste	Reported total biomedical/general waste for the day.	0-100,000 kg/day
Sharps	Reported sharps waste quantity.	0-100,000 kg/day
Infectious	Reported infectious waste quantity.	0-100,000 kg/day
Pharmaceutical	Reported pharmaceutical waste quantity.	0-100,000 kg/day
Chemical	Reported chemical waste quantity.	0-100,000 kg/day
General	Reported general waste quantity.	0-100,000 kg/day
Correctly segregated	Reported weight segregated correctly; not an external audit result.	0-100,000 kg/day
Recycled/recovered	Reported quantity recovered or recycled.	0-100,000 kg/day
Treatment/disposal completed	Reported quantity with treatment/disposal recorded.	0-100,000 kg/day
Waste handling incidents	Reported incident count.	0-10,000 incidents
Patients served	Patient count for the same reporting period, used as a denominator when available.	0-100,000 patients/day
Category/location notes	Optional user-entered context, up to 100 characters.	Text, 100 characters

SDG 12 Resource consumption

Recorded metric	Meaning and source	Validation
PPE	Reported personal protective equipment quantity.	0-1,000,000 items/day
Gloves	Reported quantity.	0-1,000,000 items/day
Masks	Reported quantity.	0-1,000,000 items/day
Syringes	Reported quantity.	0-1,000,000 items/day
Medicines	Reported medicine units; unit type is not normalized.	0-1,000,000 units/day
Packaging	Reported packaging mass.	0-100,000 kg/day
Electricity	Reported electricity use.	0-10,000,000 kWh/day
Water	Reported water use.	0-10,000,000 litres/day
Single-use items	Reported quantity.	0-1,000,000 items/day
Reusable items	Reported quantity.	0-1,000,000 items/day
Patients served	Patient count for the reporting period.	0-100,000 patients/day

SDG 12 Healthcare food waste

Recorded metric	Meaning and source	Validation
Food prepared	Reported prepared food mass.	0-100,000 kg/day
Food served	Reported served food mass.	0-100,000 kg/day
Food returned	Reported returned food mass.	0-100,000 kg/day
Food discarded	Reported discarded food mass.	0-100,000 kg/day
Meals served	Reported meal count.	0-1,000,000 meals/day
Cost of discarded food	Reported cost; currency is not normalized or specified by the app.	0-100,000,000 currency/day
Patients served	Patient count for the reporting period.	0-100,000 patients/day

SDG 12 Sustainable procurement

Recorded metric	Meaning and source	Validation
Product and supplier	User-entered product and supplier labels.	Up to 120 characters each
Product cost	User-entered cost; currency is not normalized.	0-100,000,000 currency
Packaging weight	Reported packaging mass.	0-100,000 kg
Recyclable packaging	User-reported percentage, not independently verified.	0-100%
Reusable/disposable status	User-entered product status.	Up to 30 characters
Expected waste	User-entered expected waste mass.	0-100,000 kg
Product lifespan	User-entered expected lifespan.	0-36,500 days
Transport distance	User-entered distance; route and method are not verified.	0-10,000,000 km
Supplier compliance	User-entered reported percentage; not a compliance certification.	0-100%
Treatment requirement	User-entered treatment text up to 160 characters and an optional mass field.	Text 160 characters; mass 0-100,000 kg

Calculated indicators and public-health comparisons

Indicator	Formula or comparison	Interpretation and limits
BMI	weightKg / (heightCm / 100)^2	Shown only when height and weight are both above zero. Adult screening bands are <18.5, 18.5-<25, 25-<30, and >=30. The form does not collect age; these adult bands are not suitable for children and BMI is not a diagnosis.
Moderate-equivalent activity	moderateMinutes + 2 x vigorousMinutes	Comparison uses WHO adult guidance: 150 minutes/week minimum, with 300 minutes/week associated with additional benefits; vigorous minutes are counted at 2x for this app's comparison.
Activity progress	min(100, round(equivalentMinutes / 150 x 100))	Percent of the 150 minute comparison point; not an SDG score, fitness score, or clinical outcome.
Strength progress	min(100, round(strengthDays / 2 x 100))	Percent of the 2-days/week WHO comparison point.
Sleep comparison	Recorded sleep duration compared with 7 hours	CDC general adult guidance says at least 7 hours; individual needs and recommendations vary. No sleep diagnosis is made.
Hydration completion	min(100, round(waterMl / targetMl x 100))	Shown only when the user enters a positive target. The app does not recommend or infer a universal target.
Waste per patient	totalKg / patients	Shown only when the patient count is positive and comes from the same entry/reporting period.
Segregation accuracy	segregatedCorrectlyKg / totalKg x 100	Shown only when total waste is positive. User-reported, not an inspection result. Values over total are rejected.
Disposal recorded	disposedKg / totalKg x 100	Shown only when total waste is positive. Indicates a recorded quantity, not regulatory compliance or treatment quality.

Indicator	Formula or comparison	Interpretation and limits
Tracked resources per patient	(PPE + gloves + masks + syringes) / patients	Combines only those four count fields. The units may not be comparable; it is a descriptive indicator, not a normalized efficiency score.
Food waste percentage	discardedKg / preparedKg x 100	Shown only when prepared mass is positive. Based on entered values.
Food waste per patient	discardedKg / patients	Shown only when patient count is positive.
Monthly trend	Sum of recorded values grouped by entry month	Available for waste total, food discarded, and packaging. It is the sum of entries, not a normalized daily rate; interpretation depends on reporting frequency.
Worker monitoring summary	Displays reported shift length, breaks, patients, and overtime	No fatigue threshold, medical risk score, or diagnosis is calculated.
Procurement fields	No composite score is calculated	Cost, packaging, lifespan, distance, and reported supplier values remain separate inputs. No SDG score or compliance percentage is invented.

Scientific basis and sources

Public guidance supports only the explicit comparisons listed above. The other metrics are operational data fields or arithmetic descriptions of the values a user enters. They do not use invented thresholds.

- CDC, Adult BMI Categories: https://www.cdc.gov/bmi/adult-calculator/bmi-categories.html
- WHO, Physical activity: https://www.who.int/initiatives/behealthy/physical-activity/
- CDC, About Sleep: https://www.cdc.gov/sleep/about/
- WHO, Health-care waste: https://www.who.int/news-room/fact-sheets/detail/health-care-waste

Healthcare waste labels follow the broad categories discussed by WHO (including sharps, infectious, chemical, pharmaceutical, and general waste). Local regulations and facility procedures determine actual handling, segregation, transport, and treatment; this application does not encode jurisdiction-specific compliance rules.

Technology stack and data flow

Layer	Technology and role
User interface	React 19 + TypeScript + Vite; the existing CSS theme and Lucide icons are retained. HealthcareWorkspace renders form fields from a small dataset catalogue.
Navigation and components	Existing Wouter routes, app page and Radix UI components remain. The new workspace uses the existing visual language and responsive CSS.
API	Express REST endpoints in server/_core/index.ts: GET /api/healthcare-records?dataset=..., POST /api/healthcare-records, PUT /api/healthcare-records/:id.
Authentication	Existing sdk.authenticateRequest verifies the session and supplies the server-side user openId. The browser cannot provide an owner ID. Records are queried and updated with both owner ID and record ID.
Database	MongoDB Node driver; healthcareRecords collection stores one dated document per entry with _id, userId, dataset, values, recordedAt, and source. dashboardSnapshots remains a separate legacy collection; the sample preview is no longer written to it.
Local mode	When no authenticated server session/storage is available, healthcare entries are kept in browser localStorage under bareminimum.healthcare-records.v1 and marked local-only. Sample examples are separate constants and are never persisted.
Validation	The dataset catalogue defines accepted numeric ranges and optional text lengths. The browser gives immediate range errors; the server repeats validation and rejects unknown fields, invalid values, and waste subtotals above total waste.

Implementation steps completed

- Reviewed the existing React dashboard, Express server, MongoDB helper, OAuth/session authentication, and the BareMinimum real-world guide.
- Added a typed healthcare dataset catalogue with field labels, units, validation limits, text fields, and public guidance links.
- Added owner-scoped MongoDB listing, insertion, and update functions in the existing MongoDB module.
- Added authenticated REST endpoints. Each endpoint derives the user key from the verified session, not from request input.
- Added forms, local-only fallback, loading/saving/error feedback, dated history, edit controls, empty states, separate sample examples, formulas, and selected monthly summaries.
- Updated navigation to Health & Wellness, Healthcare Worker Well-being, Healthcare Sustainability, and Trends / Reports while retaining the existing layout and theme.
- Marked the previous dashboard tiles as sample preview, stopped syncing the legacy sample snapshot to MongoDB, and kept sample examples out of personal history.
- Added this metric and implementation handoff with source links and explicit limits for unsupported claims.

Running and verifying

Configure MONGODB_URI and optionally MONGODB_DB_NAME in .env, then run the existing project command pnpm dev. The React/Vite client and Express API run together. For production, use pnpm build followed by pnpm start. pnpm check performs TypeScript validation; pnpm test runs the existing Vitest suite.

To verify the healthcare flows, use a configured MongoDB and a signed-in account: save an entry, reload its history, update it, and confirm a different account cannot read or update it. Submit values beyond the ranges and confirm HTTP 400. In a signed-out/local-only session or with the API unavailable, save and reload a local entry and confirm the status says local-only. Wearable fields are manual; real smartwatch APIs remain future integration.

Current evidence: TypeScript validation, the Vitest suite, the Vite production build, and the Express server bundle completed successfully using locally installed tools. The package-manager wrapper attempted an online bootstrap and the sandbox blocked registry access. OAuth is not configured in this environment, so signed-in MongoDB endpoint and cross-user isolation checks require a configured OAuth session; the REST routes authenticate before accessing records.

Current limits and next integration work

- No real wearable provider is connected; steps, heart rate, calories, exercise, and sleep values are manual entries.
- No healthcare facility, payroll, safety reporting, procurement, kitchen, scale, or waste-treatment system is connected. Operational data is self-reported/manual.
- Sample examples are illustrative presentation fixtures only. They do not represent a real patient, worker, facility, supplier, or measurement.
- SDG overview progress is not a single score. Read the underlying input counts and formulas; blank data means not yet recorded.
- Patient and worker roles are not currently separated by an authorization policy. Existing authenticated accounts own their own record sets; an organization-level management view requires a designed role/consent model before deployment.

Existing dashboard preview metrics

The original BareMinimum prototype still contains an illustrative interaction layer. It is not a clinical or sustainability dataset. The page displays a sample-preview notice, zero/blank placeholders, or explicit sample labels; the dashboard snapshot is retained only in browser storage and is not sent to MongoDB.

Prototype item	What it represents
Impact score / alignment score / synergy progress	Legacy arbitrary prototype values. They are not valid SDG scores or measured outcomes and are excluded from the personal-record overview.
Streak, deep work, carbon avoided, posture breaks, signal quality, team counts	Illustrative interface/demo widgets, not backed by the healthcare record database.
Activity impact points and sample activity feed	Demo content; no scientific conversion to health, worker safety, carbon, or SDG outcomes.
Deep-work boundary slider	A local UI preference with descriptive copy, not a measured workload or an evidence-based recommended limit.
Google Calendar, GitHub, Oura sample profile entries	Demo connection labels only; no provider tokens or live synchronization are present.